Integration of multiomics data with GCN to identify new cancer genes and their associated molecular mechanisms

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Integration of multiomics data with graph convolutional networks to identify new cancer genes and their associated molecular mechanisms

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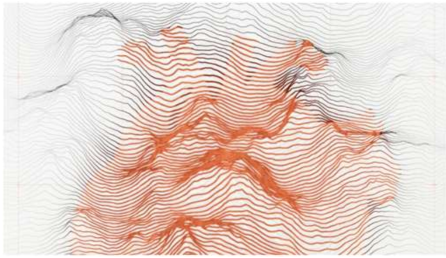

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Interpretable AI   deep learning for sequences   graph neural networks
RNA biology and therapeutics   multi-omics data integration

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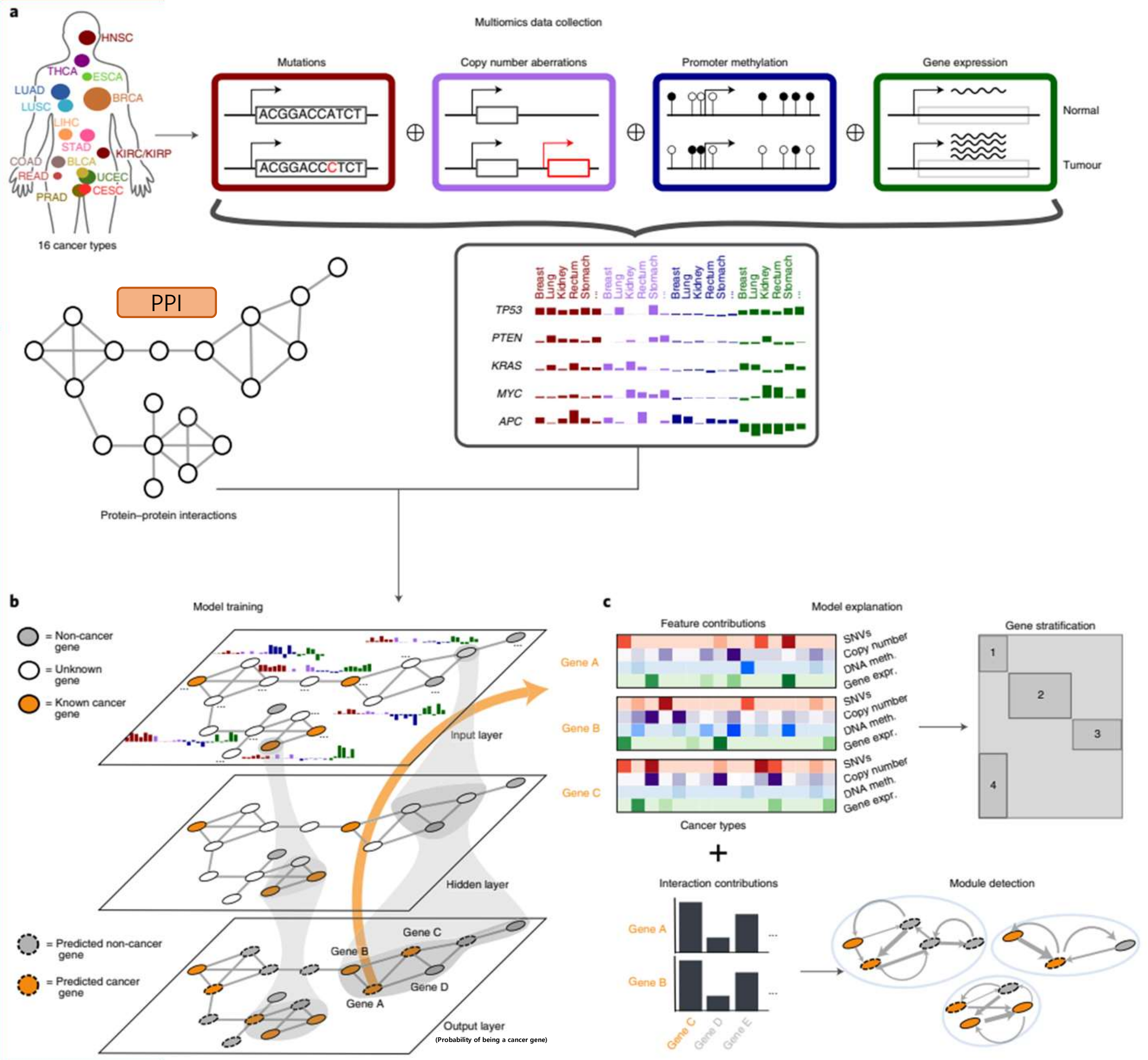
Problem

- The **explosion** of high-throughput molecular data has created significant computational hurdles in pinpointing cancer-related genes.
- Tumorigenesis is driven by both genetic and non-genetic factors; therefore, it is necessary to develop interpretable predictive models that can effectively integrate diverse types of biological data.

Method



Data

Omics data: TCGA (Pan-cancer data)

Network data: CPDB, STRING, IRefIndex, Multinet, and PCNet

Training labels: NCG, COSMIC CGC, DigSEE, KEGG, OMIM, and MutSigDB

GCN model

A deep learning model that learns node features by integrating information from each node and its neighboring nodes in a graph.

- Node=gene / Edge= PPI network
- Node feature= Multi-omics data / output : Cancer gene probability

EMOGI (Explainable Multi-omics Graph Integration)

A GCN-based model that jointly learns multi-omics data and protein–protein interaction (PPI) networks.

LRP (Layer-wise Relevance Propagation)

A technique that traces back the model's final predictions to identify and explain the specific evidence for classifying each gene as a cancer gene.

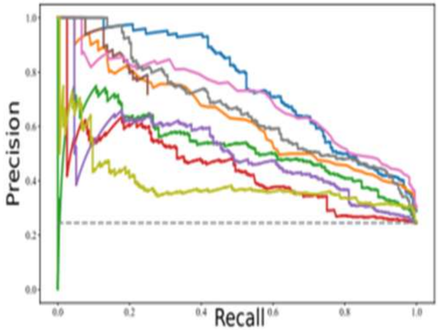
Intro

Method

Results

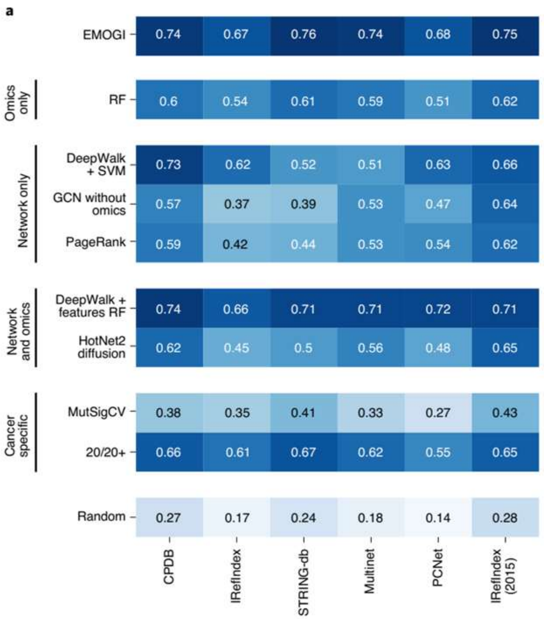
Take-home Messages

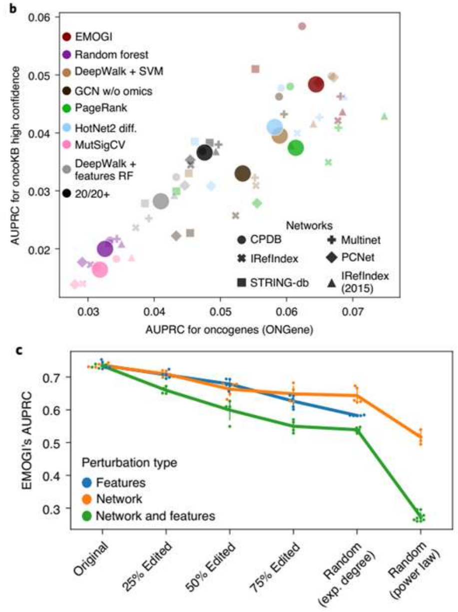
AUPRC (Area under the precision-recall curve)

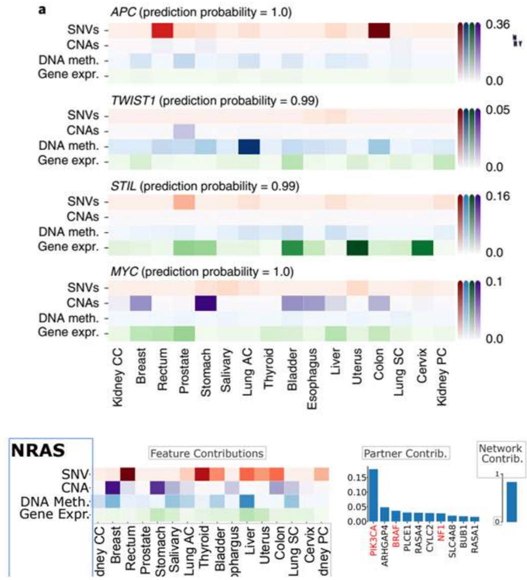


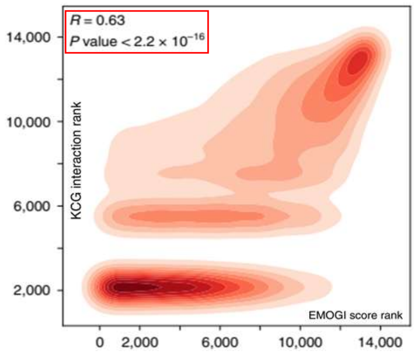
$$Precision = \frac{TP}{TP + FP} \qquad Recall = \frac{TP}{TP + FN}$$

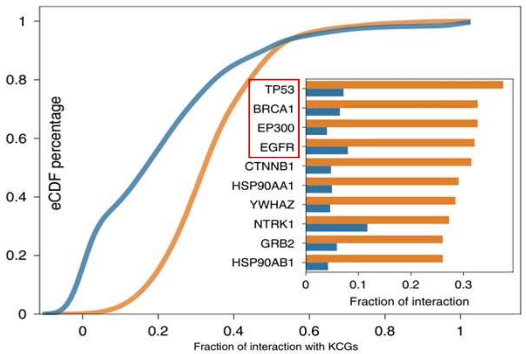
TP: True positive
FP: False Positive
FN: False Negative

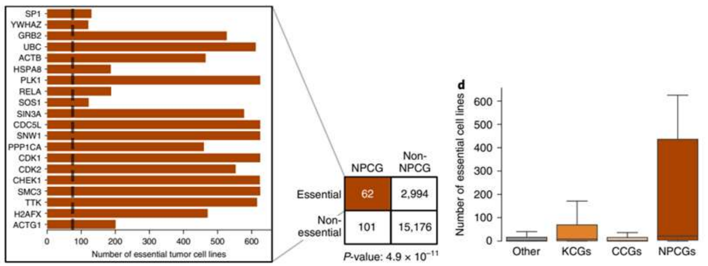


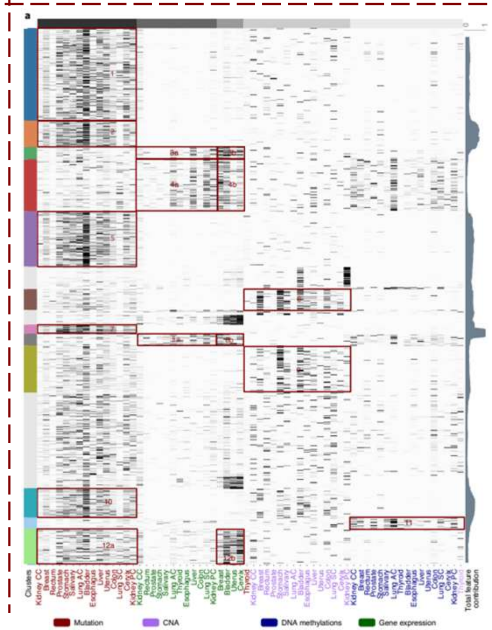


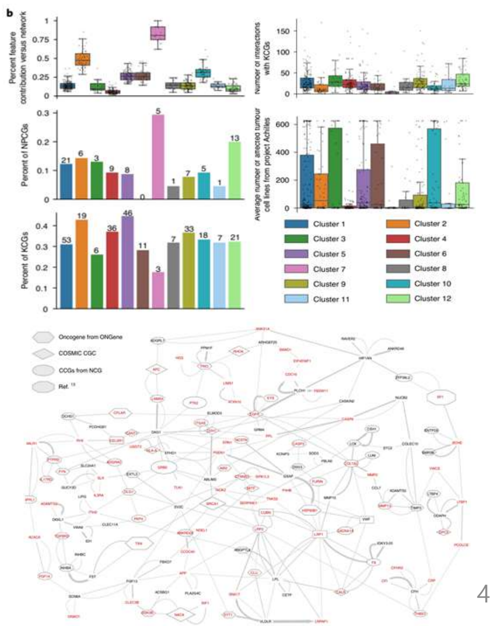












Take-home Messages

1. Comprehensive insights into cancer genes

2. Strength of multi-omics integration

3. Application of interpretable AI: LRP (Layer-wise Relevance Propagation)

4. Potential for identifying novel therapeutic targets

Conclusion

- “Multi-omics data + PPI network + Interpretable AI“ → Precision Oncology Level ↑
- It may also serve as a powerful tool for identifying causal genes in other complex diseases.

Thank you

Study

Q: LRP 의 원리가 무엇인가요?

LRP가 필요한 이유

딥러닝 모델은 보통 Black box임 (모델이 예측은 가능하지만 그 이유를 알기 어렵다. 그래서 사용하는 것이 LRP)

LRP의 핵심 아이디어

LRP는 모델의 예측값을 입력 feature 로 다시 분배하는 방법이다. 즉 prediction socre -> layer 별로 relevance 분배 -> input feature 까지 전달 -> 어떤 input이 prediction에 얼마나 기여했는지 알 수 있다.

직관적인 원리

Ex) 모델이 Fluorescence =0.9 라는 예측을 했다고 하자.

LRP는 이 0.9라는 값을 거꾸로 내려보내면서 neuron -> neuron -> input 에 각 feature 기여도를 나눈다.

결과 : '각 위치의 중요도 ' 가 나온다. (결과는 보통 heatmap으로 나타냄)

A position12 -> +0.3

G position 15 -> +0.25

T position 9 -> +0.1

수식 형태:

$$R_j = \sum_k \frac{a_j w_{jk}}{\sum_j a_j w_{jk}} R_k$$

의미

기호	의미
a_j	neuron activation
w_{jk}	weight
R_k	상위 layer relevance
R_j	하위 layer relevance

즉 activation × weight 비율로 relevance를 분배한다.

수식개념

LRP의 기본규칙은 relevance conservation. 즉, 상위 layer relevance 합 = 하위 layer relevance 합

Study

Q: Semi-supervised learning vs Self-supervised learning

Semi-supervised learning

일부 데이터만 라벨이 있고, 나머지는 라벨이 없는 경우에 사용하는 방법

목적

적은 라벨 데이터를 활용해 모델 성능을 높이는 것

Example

이미지 분류: 1,000개 이미지(라벨 있음), 100,000개 이미지(라벨 없음) → 모델이 라벨있는 데이터로 학습한 후 unlableled data의 패턴을 같이 학습.

Self-supervised learning

라벨 없이도 학습할 수 있도록 데이터를 스스로 라벨링해서 학습하는 방법

Example

BERT(**Bidirectional Encoder Representations from Transformers: 자연어 처리(NLP) 에서 문장을 이해하기 위해 사용하는 딥러닝 언어 모델)**

'문장 : the cat sat on the [MASK]' 에서 모델이 [MASK] = mat 을 예측하도록 학습하며, 라벨은 사람이 만든 것이 아닌 데이터에서 자동 생성